P0 Walk-Forward Validation

Point-in-time backtest of price-derived factor blocks | 2023-07-01 to 2025-01-31 | monthly rebalance, 19 dates | generated 2026-08-20T17:08:52

What this report does and does not establish

This is not a validation of Model 5.0. It exercises only the price-derived blocks. Quality, growth and value together carry 70% of the production score and require point-in-time fundamentals that are not yet ingested, so none of them are tested here.

It answers one question from p0.md §7E: **how much of the model's behaviour is momentum?** A momentum-only ranking that performs close to the full model would mean the fundamental blocks are decoration. Read every number below as evidence about momentum and risk, not about Model 5.0.

Two further gaps: these ablations rank **market-wide** where production ranks within sector, and momentum-only omits RS_Sector_6M_Pct (20% of the production momentum block) because the bhavcopy archive carries no sector classification.

Integrity controls

Control	Status
Universe source	Archived NSE bhavcopy per session (point-in-time, includes delisted names)
Survivorship	3384 securities; 522 delisted retained in-sample
Identifier	ISIN-bridged Security_ID; 419 face-value changes linked
Execution	Signal on close of t, fill at open of next confirmed session
Corporate actions	659 split/bonus adjusted, 11580 dividends, 620 unadjustable events excluded
Delisting policy	haircut@0.50
Costs	effective-dated Indian charges + 10bp half-spread + sqrt impact at Rs 1.00 lakh/position
Eligibility	min turnover Rs 20.00 lakh/day, trading frequency >= 0.8, history >= 200 sessions, INE-class only (ETFs excluded)
Parameters	Frozen; no re-optimisation across the window

Eligible universe per rebalance

Metric	Value
Rebalance dates	19
Mean eligible	1571
Min / max eligible	1410 / 1682
Mean considered before filters	2008

Rank information coefficient — Gross of costs

Spearman correlation between the score on the rebalance date and the forward return. Reported with its distribution because a mean carried by two extraordinary periods is a different claim from one earned consistently. **%+** is the share of periods with a positive IC.

Strategy	Horizon	Periods	Mean	Median	%+	Worst	Best
equal weight universe	1M	0	n/a	n/a	n/a	n/a	n/a
equal weight universe	3M	0	n/a	n/a	n/a	n/a	n/a
equal weight universe	6M	0	n/a	n/a	n/a	n/a	n/a
growth only	1M	19	0.0094	0.0088	58%	-0.0770	0.1777
growth only	3M	19	0.0196	0.0107	63%	-0.0978	0.1112
growth only	6M	19	0.0210	0.0339	74%	-0.0518	0.1207
model 5	1M	19	0.0453	0.0389	68%	-0.1653	0.2685
model 5	3M	19	0.0823	0.0681	89%	-0.0327	0.2191
model 5	6M	19	0.1032	0.1106	95%	-0.0723	0.2626
model 5 gated	1M	19	0.0386	0.0290	68%	-0.1749	0.2623
model 5 gated	3M	19	0.0754	0.0574	84%	-0.0285	0.2115
model 5 gated	6M	19	0.0965	0.1041	95%	-0.0702	0.2460
momentum only	1M	19	0.0346	0.0237	63%	-0.1426	0.1998
momentum only	3M	19	0.0745	0.0605	84%	-0.0244	0.2399
momentum only	6M	19	0.0959	0.0982	84%	-0.0600	0.2626
quality only	1M	19	0.0166	0.0170	58%	-0.2126	0.2553
quality only	3M	19	0.0230	0.0432	63%	-0.2169	0.2360
quality only	6M	19	0.0281	0.0771	74%	-0.2188	0.1837
random ranking	1M	19	0.0036	0.0022	53%	-0.0579	0.0569
random ranking	3M	19	-0.0007	-0.0021	42%	-0.0576	0.0519
random ranking	6M	19	0.0038	-0.0030	47%	-0.0663	0.0559
simple quality momentum	1M	19	0.0333	0.0570	63%	-0.1692	0.2098
simple quality momentum	3M	19	0.0726	0.0745	84%	-0.0198	0.2112
simple quality momentum	6M	19	0.0973	0.0878	89%	-0.0660	0.2220
value only	1M	19	0.0322	0.0347	68%	-0.0713	0.1463
value only	3M	19	0.0421	0.0388	89%	-0.0162	0.1320
value only	6M	19	0.0473	0.0427	95%	-0.0030	0.0932

Portfolio and bucket evidence — Gross of costs

vs universe is the top-20 mean period return minus the eligible universe mean, so it already controls for how the market did. **Monotonicity** is +1 when decile means fall cleanly from best score to worst, 0 when the ranking carries no ordering. **Spread** is top decile minus bottom decile.

Strategy	Horizon	Universe	Top 20	vs universe	Monotonicity	D1-D10
equal weight universe	1M	1.64%	1.64%	+0.00%	-0.164	-0.35%
equal weight universe	3M	4.15%	4.15%	+0.00%	-0.442	-1.40%

equal weight universe	6M	9.65%	9.65%	+0.00%	-0.515	-3.56%
growth only	1M	1.64%	2.84%	+1.20%	0.127	+0.06%
growth only	3M	4.15%	6.20%	+2.05%	0.248	+0.37%
growth only	6M	9.65%	15.06%	+5.42%	0.054	-0.25%
model 5	1M	1.64%	3.15%	+1.51%	0.927	+0.78%
model 5	3M	4.15%	8.65%	+4.50%	0.927	+3.95%
model 5	6M	9.65%	14.97%	+5.32%	0.952	+7.34%
model 5 gated	1M	1.64%	2.30%	+0.66%	0.806	+0.46%
model 5 gated	3M	4.15%	6.34%	+2.19%	0.770	+3.20%
model 5 gated	6M	9.65%	12.48%	+2.84%	0.891	+6.37%
momentum only	1M	1.64%	3.81%	+2.17%	0.842	+2.18%
momentum only	3M	4.15%	8.66%	+4.51%	0.964	+7.16%
momentum only	6M	9.65%	17.66%	+8.02%	0.988	+12.19%
quality only	1M	1.64%	1.63%	-0.01%	-0.345	-0.65%
quality only	3M	4.15%	4.91%	+0.76%	-0.333	-0.80%
quality only	6M	9.65%	9.12%	-0.53%	-0.333	-1.65%
random ranking	1M	1.64%	1.77%	+0.13%	0.176	+0.73%
random ranking	3M	4.15%	3.44%	-0.71%	0.103	+0.54%
random ranking	6M	9.65%	8.98%	-0.66%	0.273	+2.55%
simple quality momentum	1M	1.64%	1.76%	+0.12%	0.927	+0.91%
simple quality momentum	3M	4.15%	5.94%	+1.79%	0.988	+4.15%
simple quality momentum	6M	9.65%	11.28%	+1.64%	0.952	+7.29%
value only	1M	1.64%	3.68%	+2.05%	0.576	+0.39%
value only	3M	4.15%	10.72%	+6.57%	0.721	+1.43%
value only	6M	9.65%	24.03%	+14.38%	0.709	+2.37%

Rank information coefficient — Net of costs

Spearman correlation between the score on the rebalance date and the forward return. Reported with its distribution because a mean carried by two extraordinary periods is a different claim from one earned consistently. **%+** is the share of periods with a positive IC.

Strategy	Horizon	Periods	Mean	Median	%+	Worst	Best
equal weight universe	1M	0	n/a	n/a	n/a	n/a	n/a
equal weight universe	3M	0	n/a	n/a	n/a	n/a	n/a
equal weight universe	6M	0	n/a	n/a	n/a	n/a	n/a
growth only	1M	19	0.0229	0.0259	58%	-0.0676	0.1831
growth only	3M	19	0.0280	0.0168	68%	-0.0906	0.1168
growth only	6M	19	0.0271	0.0420	74%	-0.0485	0.1285
model 5	1M	19	0.0748	0.0681	79%	-0.1407	0.2837
model 5	3M	19	0.1001	0.0877	95%	-0.0188	0.2332

model 5	6M	19	0.1160	0.1219	95%	-0.0634	0.2770
model 5 gated	1M	19	0.0679	0.0570	79%	-0.1496	0.2779
model 5 gated	3M	19	0.0932	0.0762	95%	-0.0150	0.2253
model 5 gated	6M	19	0.1093	0.1159	95%	-0.0609	0.2604
momentum only	1M	19	0.0559	0.0476	68%	-0.1205	0.2225
momentum only	3M	19	0.0873	0.0709	89%	-0.0166	0.2465
momentum only	6M	19	0.1048	0.1074	89%	-0.0499	0.2703
quality only	1M	19	0.0421	0.0410	74%	-0.1946	0.2697
quality only	3M	19	0.0384	0.0648	74%	-0.2049	0.2489
quality only	6M	19	0.0393	0.0922	74%	-0.2107	0.1976
random ranking	1M	19	0.0043	-0.0014	47%	-0.0594	0.0584
random ranking	3M	19	-0.0002	-0.0017	42%	-0.0576	0.0496
random ranking	6M	19	0.0038	-0.0027	47%	-0.0654	0.0569
simple quality momentum	1M	19	0.0616	0.0762	68%	-0.1451	0.2358
simple quality momentum	3M	19	0.0897	0.0996	95%	-0.0058	0.2218
simple quality momentum	6M	19	0.1095	0.1094	89%	-0.0566	0.2348
value only	1M	19	0.0275	0.0313	68%	-0.0685	0.1378
value only	3M	19	0.0394	0.0361	89%	-0.0169	0.1279
value only	6M	19	0.0457	0.0427	95%	-0.0050	0.0912

Portfolio and bucket evidence — Net of costs

vs universe is the top-20 mean period return minus the eligible universe mean, so it already controls for how the market did. **Monotonicity** is +1 when decile means fall cleanly from best score to worst, 0 when the ranking carries no ordering. **Spread** is top decile minus bottom decile.

Strategy	Horizon	Universe	Top 20	vs universe	Monotonicity	D1-D10
equal weight universe	1M	-0.02%	-0.02%	+0.00%	-0.309	-0.54%
equal weight universe	3M	2.49%	2.49%	+0.00%	-0.515	-1.58%
equal weight universe	6M	7.99%	7.99%	+0.00%	-0.515	-3.73%
growth only	1M	-0.02%	1.48%	+1.50%	0.382	+0.53%
growth only	3M	2.49%	4.84%	+2.35%	0.382	+0.85%
growth only	6M	7.99%	13.70%	+5.71%	0.164	+0.24%
model 5	1M	-0.02%	1.81%	+1.84%	0.988	+1.82%
model 5	3M	2.49%	7.31%	+4.82%	0.976	+5.00%
model 5	6M	7.99%	13.61%	+5.63%	0.976	+8.39%
model 5 gated	1M	-0.02%	1.11%	+1.13%	0.927	+1.57%
model 5 gated	3M	2.49%	5.15%	+2.66%	0.927	+4.31%
model 5 gated	6M	7.99%	11.29%	+3.30%	0.891	+7.49%
momentum only	1M	-0.02%	2.27%	+2.29%	0.927	+2.83%
momentum only	3M	2.49%	7.11%	+4.62%	0.988	+7.81%
momentum only	6M	7.99%	16.11%	+8.12%	0.988	+12.83%

quality only	1M	-0.02%	0.58%	+0.60%	0.321	+0.27%
quality only	3M	2.49%	3.85%	+1.36%	0.297	+0.12%
quality only	6M	7.99%	8.06%	+0.08%	-0.273	-0.74%
random ranking	1M	-0.02%	0.10%	+0.12%	0.115	+0.78%
random ranking	3M	2.49%	1.76%	-0.73%	0.164	+0.58%
random ranking	6M	7.99%	7.31%	-0.68%	0.273	+2.60%
simple quality momentum	1M	-0.02%	0.55%	+0.58%	0.988	+1.87%
simple quality momentum	3M	2.49%	4.73%	+2.24%	0.988	+5.10%
simple quality momentum	6M	7.99%	10.04%	+2.06%	0.952	+8.24%
value only	1M	-0.02%	1.93%	+1.96%	0.770	+0.73%
value only	3M	2.49%	8.98%	+6.49%	0.746	+1.77%
value only	6M	7.99%	22.32%	+14.33%	0.746	+2.72%

Turnover

One-way turnover of a top-20 portfolio per rebalance. 1.0 means the portfolio was fully replaced. High turnover is what converts a real gross edge into no net edge.

Strategy	Mean one-way turnover	Rebalances
equal weight universe	0.066	19
growth only	0.108	19
model 5	0.295	19
model 5 gated	0.410	19
momentum only	0.492	19
quality only	0.063	19
random ranking	0.963	19
simple quality momentum	0.368	19
value only	0.163	19

Fill coverage

Share of scored positions that produced a usable forward return. A dropped position is recorded with a reason rather than silently omitted; a thin horizon here weakens every statistic above it.

Horizon	Coverage	Positions	Main reasons dropped
1M	99.6%	258705	unadjustable_action 918, no_entry_price 27
3M	99.0%	258705	unadjustable_action 2637, no_entry_price 27
6M	97.9%	258705	unadjustable_action 5436, no_entry_price 27

How to read this

The random_ranking row is the calibration. It is a seeded random score, so its IC is what "no signal" measures on this universe and horizon. Any strategy whose IC is not clearly separated from it has not demonstrated anything.

equal_weight_universe scores every name identically, so its rank IC is undefined rather than zero. It is present as the return benchmark from p0.md §7B, not as a ranking.

Judge consistency before magnitude. A mean IC of 0.03 with 70% of periods positive is stronger evidence than a mean of 0.06 driven by two periods. Check the %+ and worst columns before the mean.

Net is the number that matters. If a gross edge disappears net of costs, the ranking may still be real but is not tradable at this position size and turnover.

Known limitations of this run

Restatement bias remains. No fundamentals are used here, so it does not bite yet — but it will once the XBRL panel lands, for any period falling back to current values rather than as-filed ones.

Delisting recovery is an assumption, not an observation. The bhavcopy does not record why a security stopped trading. Re-run with --delisting-strategy zero and last_close to bound how much the headline depends on it.

Period count is modest. Monthly rebalancing over roughly four years is enough to judge cross-sectional rank IC and its stability. It is not enough to claim regime robustness across a full cycle.

Overlapping horizons are not independent. A 6M horizon sampled monthly reuses most of its window, so any t-statistic on per-period ICs overstates significance and is reported as a rough guide only.

Cost assumptions are documented estimates. Brokerage defaults to zero (discount-broker delivery); half-spread and impact coefficient are modelled, not measured. Gross and net are shown separately for that reason.

Next step

Phase B: ingest NSE XBRL filings into a point-in-time fundamental panel keyed on filing timestamp, which unlocks the quality, growth and value blocks and makes a genuine Model 5.0 walk-forward possible. Until then the fundamental 70% of the score remains untested.